

GALESBURG, IL, USA

WMO: 722089

Lat: 40.933N

Lon: 90.433W

Elev: 233

StdP: 98.56

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 94959

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.1	-17.1	-22.9	0.5	-19.1	-20.8	0.6	-16.3	12.6	-3.0	11.5	-5.0	2.6	290	0.503

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	33.0	24.0	32.1	23.6	30.6	22.8	25.8	31.0	25.0	30.0	24.1	28.8	3.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.2	19.7	28.9	23.3	18.6	27.8	22.6	17.8	27.0	81.2	31.1	76.9	29.9	73.6	29.0	32.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.9	9.2	8.2	-24.6	36.2	3.6	3.9	-27.2	39.0	-29.3	41.3	-31.3	43.5	-34.0	46.4		
(5)			-24.9	28.0	3.5	1.3	-27.4	28.9	-29.4	29.7	-31.4	30.4	-34.0	31.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.7	-4.5	-3.1	4.6	10.9	17.0	22.3	23.7	22.3	19.0	11.9	4.7	-1.6		
	DBStd	11.28	6.93	6.94	6.51	5.29	4.76	3.12	3.22	3.22	4.29	5.54	5.71	6.14		
	HDD10.0	1644	451	367	193	52	4	0	0	0	1	43	174	359		
	HDD18.3	3338	708	599	427	227	82	5	2	5	44	213	409	617		
	CDD10.0	1892	0	2	26	79	223	368	426	381	269	101	16	1		
	CDD18.3	544	0	0	2	5	42	123	169	128	63	12	0	0		
	CDH23.3	5138	0	0	18	67	394	1124	1718	1134	570	113	0	0		
	CDH26.7	1775	0	0	2	8	102	384	677	392	182	27	0	0		
(14) Wind	WSAvg	3.4	4.1	4.1	4.1	4.4	3.6	3.0	2.3	2.2	2.6	3.4	3.9	3.7		
(15) Precipitation	PrecAvg	916	42	42	55	89	125	104	102	99	77	66	66	45		
	PrecMax	1273	91	127	123	157	269	205	282	221	168	150	151	96		
	PrecMin	639	8	12	6	38	20	21	25	7	16	25	6	9		
	PrecStd	147	24	29	35	32	66	57	62	50	45	36	39	25		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.3	17.8	26.0	27.8	32.0	33.7	36.3	34.9	32.8	29.9	22.1	15.6		
		MCWB	10.1	13.0	17.3	17.2	21.3	23.8	25.0	25.1	21.9	20.9	14.3	13.8		
	2%	DB	10.0	13.2	21.5	25.4	29.1	32.3	33.6	32.5	31.0	26.7	18.0	12.1		
		MCWB	7.5	9.4	15.0	16.4	19.9	23.7	24.8	23.9	22.7	18.2	13.5	9.8		
	5%	DB	7.1	9.1	17.8	22.9	27.4	30.8	32.1	30.8	28.9	23.6	16.2	9.1		
		MCWB	5.1	6.8	13.1	15.3	18.6	22.5	24.2	23.2	21.5	17.5	12.2	7.3		
	10%	DB	4.0	6.2	14.2	20.9	25.6	28.9	30.3	28.8	27.1	21.0	13.4	7.0		
		MCWB	2.5	3.9	10.2	13.9	17.8	21.8	23.4	22.5	19.8	15.9	10.0	5.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.9	14.2	18.2	19.4	23.5	26.1	27.3	26.9	24.9	22.4	16.5	14.8		
		MCDB	13.2	16.8	24.4	24.7	29.3	31.4	32.1	32.3	30.3	27.4	19.4	16.0		
	2%	WB	7.9	10.2	15.8	17.9	21.8	24.9	26.2	25.4	23.7	20.5	14.5	10.1		
		MCDB	9.4	12.4	20.2	22.9	27.5	29.9	31.3	30.5	28.6	24.6	17.3	11.3		
	5%	WB	5.0	6.9	13.9	16.4	20.4	24.0	25.3	24.3	22.6	18.8	12.6	7.6		
		MCDB	7.1	9.0	17.5	21.6	25.0	29.0	30.5	28.7	26.9	21.6	15.5	8.9		
	10%	WB	2.7	4.1	10.7	14.8	19.2	22.9	24.4	23.3	21.5	16.8	10.3	5.0		
		MCDB	4.1	6.0	14.3	19.7	23.4	27.5	29.2	27.1	25.3	20.1	13.3	6.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.6	9.5	10.4	11.9	11.6	11.2	11.2	11.4	12.4	11.4	10.0	8.1		
		MCDBR	10.9	13.5	13.7	14.8	13.6	12.7	12.1	13.1	13.2	14.0	12.9	10.8		
	5% WB	MCWBR	8.8	9.9	8.5	8.2	6.4	6.0	5.3	5.9	6.0	7.1	8.7	8.9		
		MCWBR	10.6	12.5	12.4	13.0	11.0	10.8	10.8	11.0	11.1	11.1	11.6	10.6		
(40) Clear-Sky Solar Irradiance	taub	taub	0.289	0.307	0.337	0.387	0.419	0.426	0.443	0.427	0.394	0.348	0.303	0.290		
		taud	2.441	2.391	2.401	2.264	2.237	2.269	2.201	2.248	2.310	2.440	2.507	2.453		
	Ebn at Noon	Ebn at Noon	873	907	914	885	860	852	833	835	840	848	852	841		
		Edh at Noon	78	95	106	131	137	133	142	131	115	90	73	71		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.79	2.76	3.73	4.72	5.50	6.19	6.19	5.44	4.64	3.10	2.06	1.51		
	RadStd	RadStd	0.21	0.27	0.27	0.40	0.40	0.53	0.37	0.34	0.36	0.35	0.17	0.19		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page